Intel Core i9-14900K and Core i5-14600K

Raptor Lake Refresh makes a splash

Intel recently introduced its 14th-generation Core 'Raptor Lake Refresh' series, contending directly with AMD's Zen 4 Ryzen 7000, particularly the innovative X3D versions with 3D V-cache. The series includes three main overclockable models: the Intel Core i9-14900K at 6 GHz, priced at $589 (around INR 57,869), the $409 Core i7-14700K, and the $319 Core i5-1600K. Available from October 17, these models are competitively priced, aligning with the cost-conscious Indian market, and promoting better adoption. Intel also offers a more affordable KF-series without integrated graphics.

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Performance 89
Value for Money....75

These processors are an evolution of the 13th-Gen Raptor Lake chips. The 13th generation initially bested AMD's Ryzen 7000, but the latter's 7000X3D models soon edged ahead in gaming performance. Now, Intel suggests its 14th-Gen would gain the crown back.

Diving deeper into the 14th Gen, also codenamed Raptor Lake Refresh, the flagship Core i9-14900K boasts 24 Cores/32 Threads, with a 6 GHz peak frequency. Embracing the hybrid architecture from the 12th Gen Alder Lake, these processors remain compatible with Intel's 600 and 700 series motherboards, requiring only a BIOS update. Let's take a closer look at the two processors that we're reviewing today.

INTEL 14TH GEN CORE PROCESSOR SPECIFICATIONS

Intel's 14th Gen desktop processors, known as Raptor Lake Refresh, bring a refined architecture and enhanced performance capabilities. With an initial release of six processors, Intel aims to cater to a broad spectrum of users, from gamers to professionals.

Leading the lineup is the flagship Core i9-14900K, a powerhouse with eight Performance-Cores (P-Cores) and sixteen Efficiency-Cores (E-Cores). Its unique design permits an astounding 5.6 GHz Turbo frequency on P-Cores and 4.4 GHz on E-Cores. Notably, there's a variant, the 14900KF, that mirrors the same performance specs but comes without an integrated GPU.

The mid-range segment sees the introduction of the Core i7-14700K and 14700KF. These chips offer eight P-Cores and twelve E-Cores, with P-Cores capable of reaching 5.5 GHz Turbo frequency and E-Cores maxing out at 4.3 GHz.

For the mainstream user, the Core i5-14600K and 14600KF are available, boasting six P-Cores and eight E-Cores. These chips can reach Turbo frequencies of 5.3 GHz and 4.0 GHz for P-Cores and E-Cores, respectively.

The graphical capabilities of these processors remain consistent with the previous generation, leveraging Intel UHD Graphics 770. Importantly, for those considering upgrades, the 14th Gen retains compatibility with the LGA1700 socket, ensuring continuity from the 12th and 13th Gen processors. This strategic decision facilitates smoother transitions for users and motherboard manufacturers.

PERFORMANCE

Cinebench R23

Starting off with the usual stuff, we see the new Intel Core i9-14900K score 37913 in the multi-threaded run and 2299 points in the single-threaded run. Compared to the AMD Ryzen 9 7950X which scored 37950 in the multi-threaded run and 1997 in the single-threaded run, we've got a tie on one front (nT) and a significant lead in single-threaded performance. Single-threaded performance has a much larger impact overall so Intel seems to be off to a good start already.

Blender

Then we have Blender which is a fairly well-known open-source rendering software used by movie studios as well as by game developers. There are standard benchmarking scenes on Blender and we use multiple iterations of the Classroom, Monster, and Junkshop scenes for our tests. We see the 14900K come really close to the Ryzen 9 7950X3D but lagging behind by about 1 per cent. The 14600K on the other hand comes very close to the 12900K and bests the 13600K by a small margin.